

Abel Prasad

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EDUCATION

Pennsylvania State University

Expected December 2026

B.S. Computer Science, Abington PA

Coursework: Data Structures and Algorithms, Operating Systems, Databases, Machine Learning, AI, Data Mining, Software Testing

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS

Backend: FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, Resend, Upstash Redis

Frontend: React, Next.js 15, Tailwind CSS

AI/ML: LLM Orchestration, Agentic AI, RAG, NLP, Groq, Ollama, spaCy

Databases: PostgreSQL, MongoDB, SQLite, Supabase

Infra: Docker, Linux, Git, Playwright, Vercel, Sentry, Telegram Bot API

EXPERIENCE

Software Engineer Intern

Feb 2026 - Present

DOGSRUN

- Migrated the platform from Microsoft Power Pages/Dataverse to a production Next.js 15 + Supabase stack; live at dogsrun.org with 150+ real shelter dogs, automated breed/age/weight matching, and transactional email via Resend.
- Architected the full auth and routing system - centralized session context, three isolated route trees, role-based redirects, and a standalone admin portal with org approval workflow and dog management.
- Hardened the API with Upstash Redis rate limiting, HTML injection prevention, mass-assignment protection, and a 45-test Playwright e2e suite covering auth and security regressions.

Software Engineer

Dec 2025 - Present

Scout - Autonomous Internship Agent

- Built a self-hosted multi-agent system that automatically finds, scores, and tracks internship postings - LLM planning handles goal decomposition, Playwright handles scraping, FastAPI + SQLite handle everything else.
- Cut manual review time by 80% by rewriting the scoring engine with role/skill keyword matching and negative filters - went from skimming 50+ noisy results daily to a tight shortlist of relevant postings.
- Delivers a ranked digest every morning via HTML email and Telegram with each posting's score, match rationale, and apply link - monitors 6+ job boards daily and surfaces only the top-scored results.

PROJECTS

Fillr

Oct 2025 - Present

AI Job Application Chrome Extension

- Reduced time-per-application by ~70% by building a Manifest V3 Chrome Extension with a 40+ field detection engine; works across 90%+ of ATS platforms including Greenhouse, Lever, Workday, and Taleo.
- Integrated Groq's Llama 3.3 70B for runtime cover letter and Q&A; generation - the DOM parser handles text inputs, dropdowns, checkboxes, and file uploads across dynamically rendered SPAs where most autofill tools fail.

FanTravels

2025

Full-Stack Web App, 5-person team

- Designed the PostgreSQL schema for a platform mapping pop culture franchises to UNESCO World Heritage Sites - defined tables, relationships, and foreign keys to model many-to-many connections between media properties, characters, and locations.
- Built the FastAPI backend with RESTful endpoints serving the Next.js/TypeScript frontend; handled all data modeling decisions and query design so the team could build features against a stable, well-structured API.

Mini-Pupper Robotics Capstone

Jan 2026 - May 2026

CMPSC 488, Penn State Abington

- Locked down the robot's control network by implementing mTLS across all microservices - every inter-service call is authenticated and encrypted, which became critical once the system moved off localhost onto a shared network.
- Cut debugging time by ~50% by building a MongoDB telemetry pipeline and Redis state layer; the team can inspect live sensor readings and nav decisions in real time instead of adding print statements and redeploying.
- Integrated AprilTag detection with a ROS2 Python control layer for landmark-based localization - the robot navigates a maze autonomously without GPS or manual waypoints.